Data and Moments

Companion Specification for the Segmented Search Model

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1 Purpose of this document

This document is the specification sheet for the data-cleaning pipeline. It is meant to be handed—together with the codebase—to a programmer (or to a future version of the author) so that the pipeline can be built without ambiguity.

It follows the paper's structure: first the data sources and what we extract from each (mirroring the paper's Data section), then the moments and their identification role (mirroring the paper's Moments and Identification section), then the estimation windows and the multi-regime logic (mirroring the paper's methodology sections). Moment construction recipes, influence functions, and the file layout close the document.

Convention. *Unskilled*: no completed four-year degree. *Skilled*: bachelor's degree or higher. Formally, $\text{skilled}_i = \mathbf{1}\{\texttt{EDUC}_i \geq 111\}$ (IPUMS code 111 = Bachelor's degree).

Part I Data Sources

2 Current Population Survey — Basic Monthly

What it gives us. Unemployment rates, skilled share, training share, all transition rates (UE, EU, EE), and the industry composition needed for JOLTS vacancy allocation.

Source. IPUMS-CPS Basic Monthly extracts (https://cps.ipums.org).

Extract window. January 2001 – December 2022. One long file covering all estimation windows plus a buffer for panel matching.

Required variables.

Panel matching for transitions. The CPS uses a 4-8-4 rotation. Match on `CPSIDP` across consecutive months within a rotation spell. Valid month-pairs: MISH $\in \{1\text{–}3,\ 5\text{–}7\}$ matched to the following month.

For the skilled EE rate, the January 2004+ CPS redesign added an employer-change question; alternatively, use industry/occupation changes across matched months as a proxy.

COVID-era correction. During March–June 2020, the CPS suffered response-rate drops and a well-documented misclassification of temporarily laid-off workers as "employed, absent from work." Apply the BLS correction: reclassify persons coded employed-absent who report reason "other" and are in industries with mass layoffs as unemployed on temporary layoff.

3 CPS Annual Social and Economic Supplement (ASEC)

What it gives us. All wage moments: means, medians, standard deviations, variances, third central moments, log skill premium.

Source. IPUMS-CPS ASEC samples.

Extract window. ASEC 2003–2022 (covering income earned in 2002–2021).

Required variables.

Table 1: CPS Basic Monthly — required variables.

IPUMS name	Type	Use
`YEAR`	id	Calendar year
`MONTH`	id	Calendar month
`CPSID`	id	Household linking key
`CPSIDP`	id	Person linking key (cross-month matching)
`MISH`	id	Month-in-sample (rotation group 1–8)
`WTFINL`	wt	Basic Monthly final weight
`EMPSTAT`	stat	Employment status (employed / unemployed / NILF)
`LABFORCE`	stat	In labour force (yes/no)
`EDUC`	demog	Educational attainment (detailed). Skill indicator: $S = 1$ iff `EDUC` ≥ 111.
`SCHLCOLL`	demog	Currently enrolled in school/college
`AGE`	demog	Age (restrict to 16–64)
`SEX`	demog	Sex
`RACE`	demog	Race (for robustness / sub-samples)
`IND`	occ	Industry (Census codes $\rightarrow$ 2-digit NAICS). Used for JOLTS vacancy allocation.
`CLASSWKR`	stat	Class of worker (wage/salary vs. self-employed)

Table 2: CPS ASEC — required variables.

IPUMS name	Type	Use
`YEAR`	id	Survey year
`CPSID`	id	Household linking
`CPSIDP`	id	Person linking
`ASECWT`	wt	ASEC supplement weight
`AGE`	demog	Age (restrict 16–64)
`SEX`	demog	Sex
`EDUC`	demog	Educational attainment
`EMPSTAT`	stat	Employment status
`INCWAGE`	income	Annual wage/salary income
`WKSWORK1`	income	Weeks worked last year (continuous)
`UHRSWORKLY`	income	Usual hours per week last year
`CLASSWKR`	stat	Class of worker
`CPI99`	price	CPI-U deflator (or deflate with BLS CPI-U-RS)

Hourly wage construction.

$$w_{it} = \frac{\texttt{INCWAGE}_{it}}{\texttt{WKSWORK1}_{it} \times \texttt{UHRSWORKLY}_{it}} \times \frac{\text{CPI-U-RS}_{2012}}{\text{CPI-U-RS}_t}. \tag{1}$$

Trim at the 1st and 99th percentiles of the pooled within-window distribution.

Sample restrictions (ASEC).

1. Age 16–64 at survey date.
2. Employed wage/salary workers (`CLASSWKR`).
3. Positive wage income, weeks, and hours.
4. Exclude self-employed and unpaid family workers.

4 Job Openings and Labor Turnover Survey (JOLTS)

What it gives us. Vacancy counts for market tightness θ_U and θ_S.

Source. BLS JOLTS (https://www.bls.gov/jlt/).

Extract window. December 2000 – December 2022.

Required series. Total nonfarm job openings (SA, level): `JTS000000000000000JOL`. Openings by major industry (2-digit NAICS): `JTS{industry}00000000JOL` for each published group.

Vacancy allocation to skill segments. JOLTS does not disaggregate by education requirement. Allocation method:

1. From CPS Basic Monthly, compute the share of employed skilled workers in each 2-digit NAICS industry k within the relevant window: $s_{k,w} = E_{S,k,w}/(E_{U,k,w} + E_{S,k,w})$.
2. From JOLTS, obtain monthly openings by industry k: $V_{k,t}$.
3. Skilled vacancies: $V_{S,t} = \sum_k s_{k,w} V_{k,t}$. Unskilled: $V_{U,t} = \sum_k (1 - s_{k,w}) V_{k,t}$.

The shares $s_{k,w}$ are window-specific.

Part II Moments and Identification

5 Moment catalogue

Table 3 lists all 25 targeted moments. Each is computed separately for every estimation window.

Table 3: Complete list of targeted moments.

Code name	Block	Definition
`ur_total`	Stocks	Overall unemployment rate: $(U_U + U_S)/(N_U + N_S)$.
`ur_U`	Stocks	Unskilled unemployment rate: U_U/N_U.
`ur_S`	Stocks	Skilled unemployment rate: U_S/N_S.
`skilled_share`	Stocks	Skilled share of labour force: $N_S/(N_U + N_S)$.
`training_share`	Stocks	Currently enrolled in college without BA, as a fraction of the labour force.
`emp_var_U`	Stocks	Variance of real hourly wages, employed unskilled.
`emp_cm3_U`	Stocks	Third central moment of wages, employed unskilled.
`emp_var_S`	Stocks	Variance of real hourly wages, employed skilled.

Code name	Block	Definition
`emp_cm3_S`	Stocks	Third central moment of wages, employed skilled.
`jfr_U`	Trans	Monthly unskilled job-finding rate: $\Pr(E_{t+1} \mid U_t, U)$.
`sep_rate_U`	Trans	Monthly unskilled EU separation rate: $\Pr(U_{t+1} \mid E_t, U)$.
`jfr_S`	Trans	Monthly skilled job-finding rate: $\Pr(E_{t+1} \mid U_t, S)$.
`sep_rate_S`	Trans	Monthly skilled EU separation rate: $\Pr(U_{t+1} \mid E_t, S)$.
`ee_rate_S`	Trans	Monthly skilled employer-to-employer rate. Pins down job-ladder intensity.
`training_rate`	Trans	Flow into training per unskilled unemployed per quarter.
`mean_wage_U`	Wages	Mean real hourly wage, employed unskilled (normalised).
`mean_wage_S`	Wages	Mean real hourly wage, employed skilled (normalised).
`p50_wage_U`	Wages	Median real hourly wage, employed unskilled.
`p50_wage_S`	Wages	Median real hourly wage, employed skilled.
`wage_premium`	Wages	Log skill premium: $E[\log w \mid S] - E[\log w \mid U]$.
`wage_sd_U`	Wages	Std. dev. of wages, employed unskilled.
`wage_sd_S`	Wages	Std. dev. of wages, employed skilled.
`theta_U`	Tight	Unskilled V-U ratio: $\theta_U = V_U/U_U$.
`theta_S`	Tight	Skilled V-U ratio: $\theta_S = V_S/U_S$.

6 Identification

This section explains *what each moment identifies* in the model. The purpose is to make clear why each moment is in the target list and which parameters it primarily constrains.

6.1 Stock moments

- `ur_U`, `ur_S`: pin down the balance between job-finding and separation rates in each segment. Given tightness, they constrain matching efficiencies and shock hazards.
- `skilled_share`: constrains the training cutoff $\bar{x}(z)$ and therefore the interaction between the type distribution (a_ℓ, b_ℓ) and the training cost (a_c).
- `training_share`: directly identifies the mass in the training state, hence the cutoff $\bar{x}$ conditional on the training-completion rate ϕ.
- `emp_var_U`, `emp_cm3_U`: higher-order moments of the unskilled wage distribution. Because the unskilled wage is linear in match quality p (see model notes eq. 12), the variance and skewness of wages map into the shape of the match-quality distribution. Specifically, `emp_var_U` identifies the damage-shock shape α_U and `emp_cm3_U` helps separate it from the reservation threshold p^*.
- `emp_var_S`, `emp_cm3_S`: analogous role for the skilled block. Constrain the offer distribution (a_Γ, b_Γ) and the skilled separation cutoff p^*_S.

6.2 Transition rates

- `jfr_U`, `jfr_S`: the job-finding rate in each segment is $f_j = \theta_j q_j(\theta_j)$. Given tightness from the JOLTS moments, these identify matching efficiencies μ_U, μ_S and elasticities η_U, η_S.
- `sep_rate_U`, `sep_rate_S`: the EU flow. In the unskilled block this is $\lambda_U G_z(p^*) + \nu$; in the skilled block it combines exogenous separations ξ_S, endogenous destruction $\lambda_S \Gamma_z(p_S^*)$, and demographic exit. These moments identify the shock arrival rates and, jointly with the wage-dispersion moments, the shape of quality shocks.
- `ee_rate_S`: the skilled job-to-job rate. In the model this is driven by the OJS hazard $\theta_S q_S(\theta_S)(1 - \Gamma_z(p))$ integrated over the searcher distribution. This is the key moment separating the offer distribution Γ_z from p_S^*, and also pins down the OJS cost σ_S (since workers search only if the gain exceeds σ_S).
- `training_rate`: together with `training_share`, pins down the flow into training and hence the cutoff $\bar{x}(z)$ and cost function $c(x)$.

6.3 Wages

- `mean_wage_U`, `mean_wage_S`: wage levels identify the productivity shifters P_U, P_S given the bargaining weights.
- `wage_premium`: the log skill premium is the primary moment identifying the *return to training*. It constrains the ratio P_S/P_U and the value of the training option.
- `p50_wage_U`, `p50_wage_S`: the medians, together with the means, pin down the asymmetry of the wage distribution within each skill group.
- `wage_sd_U`, `wage_sd_S`: dispersion identifies bargaining weights β_U, β_S. Higher bargaining power compresses wages; lower power allows more pass-through of match-quality heterogeneity.

6.4 Tightness

- `theta_U`, `theta_S`: directly constrain vacancy costs k_U, k_S through free entry. Given matching parameters and the composition of unemployed workers, free entry pins down tightness uniquely.

6.5 Deep vs. regime-specific identification

Deep parameters are identified primarily from "structural" moments whose cross-regime variation is limited: matching elasticities from the *shape* of the JFR–tightness relationship across segments; matching efficiencies from the *level* of job-finding rates given tightness (these are technological, not regime-dependent); bargaining weights from wage dispersion; the type distribution from the skilled share and wage premium; the training-cost coefficient from training flows; and flow payoffs b_U, b_T, b_S from the level of unemployment values and training incentives. The discount rate r and training completion rate ϕ are calibrated externally; the demographic turnover rate ν is pre-estimated from CPS labour-force exit hazards (see Part VI).

Regime-specific parameters are identified from moments that shift across windows: productivity shifters from wage levels, vacancy costs from tightness levels, shock arrival rates and damage/offer shapes from separation rates and wage-dispersion shifts, and exogenous skilled separation from the skilled EU rate.

Part III Estimation Windows

7 The four windows

The multi-regime estimation strategy (see the paper plan) requires moments computed over four distinct periods, each approximating a stationary regime:

Table 4: Estimation windows.

Label	Code	Period	Role
Pre-FC baseline	base_fc	Jan 2003 – Dec 2007	Stage 1: estimate all parameters.
Financial crisis	crisis_fc	Jan 2008 – Jun 2009	Stage 2a: regime-specific only.
Pre-COVID baseline	base_covid	Jan 2015 – Dec 2019	Structural stability check + 2nd baseline.
COVID crisis	crisis_covid	Mar 2020 – Dec 2021	Stage 2b: regime-specific only.

Each window produces 25 moment values plus a 25×25 variance-covariance matrix from influence functions.

7.1 Common sample restrictions

1. Age 16–64, civilian labour force.
2. Exclude armed forces.
3. Wage moments: wage/salary workers, positive income/weeks/hours, exclude self-employed.
4. Transition rates: matched month-pairs with valid `CPSIDP`, MISH $\in \{1\text{–}3,\ 5\text{–}7\}$.

7.2 Window-specific notes

base_fc (Jan 2003 – Dec 2007). Starts two years after the 2001 trough. NBER expansion: Nov 2001 – Dec 2007.

crisis_fc (Jan 2008 – Jun 2009). NBER contraction: Dec 2007 – Jun 2009. Full contraction included.

base_covid (Jan 2015 – Dec 2019). Mature post-crisis expansion. Starting in 2015 avoids the early recovery (2010–14). Tests structural stability of deep parameters across a decade.

crisis_covid (Mar 2020 – Dec 2021). NBER peak Feb 2020. Extended through Dec 2021 to capture the full reallocation period. BLS misclassification correction applied for Mar–Jun 2020.

7.3 ASEC year mapping

Since ASEC reports income for the *previous* calendar year, the ASEC surveys used for each window are:

- base_fc: ASEC 2004–2008 (income years 2003–2007).
- crisis_fc: ASEC 2009–2010 (income years 2008–2009).
- base_covid: ASEC 2016–2020 (income years 2015–2019).
- crisis_covid: ASEC 2021–2022 (income years 2020–2021).

7.4 Stationarity caveat

The model produces stationary-equilibrium moments. Short windows (especially the 18-month financial crisis) are not true stationary states. We interpret crisis-window averages as the "quasi-stationary" centre of gravity. Robustness to window endpoints will be checked.

Part IV Moment Construction Recipes

8 Labour-market stocks

All recipes below are applied identically within each window w.

`ur_total`, `ur_U`, `ur_S`. For each month t in window w:

$$\mathrm{ur}_{j,t} = \frac{\sum_{i:\,\mathrm{unemp},\,j} \mathtt{WTFINL}_i}{\sum_{i:\,\mathrm{LF},\,j} \mathtt{WTFINL}_i}, \qquad j \in \{U, S\}.$$

Average over all months in the window.

`skilled_share`. Monthly fraction of the labour force that is skilled, averaged over the window:

$$\mathtt{skilled_share}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i:\,\mathrm{LF},\,S} \mathtt{WTFINL}_{i,t}}{\sum_{i:\,\mathrm{LF}} \mathtt{WTFINL}_{i,t}}.$$

`training_share`. Among all persons 16–64 in the labour force or enrolled: identify those enrolled in college ($\mathtt{SCHLCOLL} \in \{4, 5\}$) without a BA ($\mathtt{EDUC} < 111$). Ratio (weighted) to total. Averaged monthly.

9 Transition rates

`jfr_U`, `jfr_S`. From matched CPS month-pairs $(t, t{+}1)$:

$$\mathrm{jfr}_{j,t} = \frac{\sum_{i:\,U_t \to E_{t+1},\,j} \mathtt{WTFINL}_{i,t}}{\sum_{i:\,U_t,\,j} \mathtt{WTFINL}_{i,t}}.$$

Averaged over all valid month-pairs in the window.

`sep_rate_U`, `sep_rate_S`.

$$\mathrm{sep}_{j,t} = \frac{\sum_{i:\,E_t \to U_{t+1},\,j} \mathtt{WTFINL}_{i,t}}{\sum_{i:\,E_t,\,j} \mathtt{WTFINL}_{i,t}}.$$

This is the $E \to U$ flow only. Job-to-job moves are not separations.

`ee_rate_S`. Among matched skilled employed workers in both t and $t+1$, identify those with a different employer (CPS employer-change question, available from Jan 2004): $\mathrm{ee}_{S,t} = \Pr(\text{different employer}_{t+1} \mid E_t, S)$. Fallback: industry/occupation code change as a proxy.

`training_rate`. Baseline (cross-sectional proxy):

$$\mathtt{training_rate}_w \approx \frac{\mathtt{training_share}_w}{\mathtt{training_share}_w + \mathrm{ur}_{U,w}}.$$

10 Wages

All from the ASEC. Hourly wages from (1), trimmed at 1st/99th percentiles within the window.

Normalisation. Each window is normalised by its own pooled median hourly wage $\tilde{w}_{50,w}$: $w_{it}^{\text{norm}} = w_{it}/\tilde{w}_{50,w}$.

`mean_wage_U`, `mean_wage_S`. Weighted mean of w^{norm} by skill group, pooled across ASEC years in the window.

`p50_wage_U`, `p50_wage_S`. Weighted median.

`wage_premium`. $E[\log w^{\text{norm}} \mid S] - E[\log w^{\text{norm}} \mid U]$. (Invariant to normalisation.)

`wage_sd_U`, `wage_sd_S`. Weighted standard deviation.

`emp_var_U`, `emp_var_S`. $= (\texttt{wage_sd_j})^2$.

`emp_cm3_U`, `emp_cm3_S`.

$$\hat{\mu}_3^j = \frac{\sum_{i \in j} w_i \, (w_i^{\text{norm}} - \bar{w}_j)^3}{\sum_{i \in j} w_i}.$$

11 Market tightness

Step 1. Unemployment by skill from CPS Basic: $U_{j,t}$.

Step 2. Vacancies by skill via industry allocation (Section 4): $V_{S,t} = \sum_k s_{k,w} \, V_{k,t}$, $V_{U,t} = \sum_k (1 - s_{k,w}) \, V_{k,t}$.

Part V SMM Weights, Influence Functions, and File Layout

12 SMM weight matrix

The SMM objective for window w:

$$Q_w(\theta) = [\hat{m}_w - m(\theta)]' \, \hat{\Xi}_w \, [\hat{m}_w - m(\theta)].$$

Diagonal initial pass: $\hat{\Xi}_w = \text{diag}(1/\hat{\sigma}_{1,w}^2, \ldots, 1/\hat{\sigma}_{K,w}^2)$. Optimal second pass: $\hat{\Xi}_w = \hat{\Sigma}_w^{-1}$.

13 Influence functions

Each moment $\hat{m}_k = \frac{1}{N} \sum_i \psi_k(z_i) + o_p(N^{-1/2})$. The variance-covariance matrix: $\hat{\Sigma}_w = \frac{1}{N_w} \sum_i \boldsymbol{\psi}(z_i) \, \boldsymbol{\psi}(z_i)'$.

Sample mean. $\psi_i = y_i - \hat{m}$.

Sample variance. $\psi_i = (y_i - \bar{y})^2 - \hat{\sigma}^2$.

Third central moment. $\psi_i = (y_i - \bar{y})^3 - \hat{\mu}_3 - 3\hat{\sigma}^2(y_i - \bar{y})$.

Ratio / rate. $\psi_i = (y_i - \hat{r}\, x_i)/\bar{x}$.

Median. $\psi_i = \frac{1}{2f(\hat{m}_{50})}(\mathbf{1}\{y_i \le \hat{m}_{50}\} - 0.5)$.

Implementation. For each window: (i) compute $\hat{m}_w$; (ii) compute $\boldsymbol{\psi}(z_i)$ for each observation; (iii) form $\hat{\Sigma}_w$; (iv) store both $\hat{m}_w$ and $\hat{\Sigma}_w$.

14 File and directory layout

```
project_root/
  code/
    solver/                  # model solver (existing)
      model_main.jl
      params.jl, grids.jl, unskilled.jl, skilled.jl
      solver.jl, equilibrium.jl, plots.jl
    smm/                     # SMM estimation (existing)
      smm_main.jl, moments.jl, smm_params.jl, smm.jl
    data_cleaning/           # NEW
      clean_cps_basic.jl
      clean_cps_asec.jl
      clean_jolts.jl
      make_transitions.jl
      make_moments.jl        # master: 25 moments x 4 windows
      utils.jl               # weighting, IFs, trimming
  data/
    raw/
      cps_basic/             # IPUMS extract
      cps_asec/              # IPUMS ASEC extract
      jolts/                 # BLS flat files
    derived/
      cps_basic_clean.arrow
      cps_asec_clean.arrow
      transitions.arrow
      jolts_clean.arrow
      moments_base_fc.csv
      moments_crisis_fc.csv
      moments_base_covid.csv
      moments_crisis_covid.csv
      sigma_base_fc.csv     # 25x25 var-cov
      sigma_crisis_fc.csv
      sigma_base_covid.csv
      sigma_crisis_covid.csv
  output/
    plots/
    tables/
```

15 Summary: moment-to-data mapping

Table 5: Moment-to-data mapping.

Moment	Source	Key variables
`ur_total`	CPS Basic	`EMPSTAT`, `EDUC`, `WTFINL`
`ur_U`	CPS Basic	same
`ur_S`	CPS Basic	same
`skilled_share`	CPS Basic	`EDUC`, `LABFORCE`, `WTFINL`
`training_share`	CPS Basic	`SCHLCOLL`, `EDUC`, `WTFINL`
`emp_var_U`	CPS ASEC	`INCWAGE`, `WKSWORK1`, `UHRSWORKLY`, `EDUC`
`emp_cm3_U`	CPS ASEC	same
`emp_var_S`	CPS ASEC	same
`emp_cm3_S`	CPS ASEC	same
`jfr_U`	CPS Basic (matched)	`CPSIDP`, `EMPSTAT`, `EDUC`
`jfr_S`	CPS Basic (matched)	same
`sep_rate_U`	CPS Basic (matched)	same
`sep_rate_S`	CPS Basic (matched)	same
`ee_rate_S`	CPS Basic (matched)	`CPSIDP`, `EMPSTAT`, `EDUC`, employer ID
`training_rate`	CPS Basic	`SCHLCOLL`, `EMPSTAT`, `EDUC`
`mean_wage_U`	CPS ASEC	`INCWAGE`, `WKSWORK1`, `UHRSWORKLY`, `EDUC`
`mean_wage_S`	CPS ASEC	same
`p50_wage_U`	CPS ASEC	same
`p50_wage_S`	CPS ASEC	same
`wage_premium`	CPS ASEC	same
`wage_sd_U`	CPS ASEC	same
`wage_sd_S`	CPS ASEC	same
`theta_U`	JOLTS + CPS	openings by industry, `IND`, `EDUC`
`theta_S`	JOLTS + CPS	same

Part VI Externally Set Parameters

16 Overview

Three parameters are set outside the SMM estimation loop. Two are *calibrated* from external data sources; one is *pre-estimated* from CPS transition data using a hazard-function approach. All three are held fixed throughout all estimation stages.

Table 6: Externally set parameters.

Symbol	Name	Value	Frequency	Source / method
Calibrated				
r	Discount rate	0.0033	monthly	4% annual real rate (risk-free rate)
ϕ	Training completion rate	0.0208	monthly	NCES/IPEDS time-to-degree: 4-year BA
Pre-estimated from data				
ν	Demographic turnover	—	monthly	CPS Basic Monthly hazard function

17 Calibrated parameters

17.1 Discount rate r

The discount rate enters every value function in the model as a level shifter: for a generic HJB of the form $(r + \nu + \cdots)V = \ldots$, a higher r uniformly lowers continuation values. But this effect is almost perfectly collinear with changes in the flow payoffs b_U, b_T, b_S and the productivity shifters P_U, P_S—raising r by ϵ and simultaneously adjusting flow payoffs can leave all observable moments nearly unchanged.

Formally, the Jacobian $\partial m/\partial r$ (evaluated at the baseline) is close to a linear combination of $\partial m/\partial b_U$, $\partial m/\partial b_S$, and $\partial m/\partial P_U$, making the Hessian of the SMM objective near-singular when r is included. This is a standard identification failure: the discount rate has no moment that pins it down independently of flow payoffs. Every quantitative search model in the literature calibrates r externally (typically to match a risk-free rate), and we follow that convention.

17.2 Training completion rate ϕ

The training completion rate ϕ governs the duration of the training spell: expected duration is $1/\phi$ months (ignoring demographic exit, which shortens it to $1/(\phi + \nu)$).

In principle, ϕ is jointly identified by the training share and the training flow rate. In steady state, training_share $\approx$ training_flow$/(\phi+\nu)$, so given the two moments one could back out ϕ. In practice, however, the training flow rate is itself a rough cross-sectional proxy (see Part IV), and small measurement errors in the training share translate into large errors in ϕ because the training share is small ($\approx 2\%$). The implied ϕ would be extremely noisy.

More fundamentally, ϕ interacts tightly with the training cutoff $\bar{x}(z)$: a model with high ϕ (fast completion) and a low cutoff (few entrants) can produce similar training stocks and flows as a model with low ϕ and a higher cutoff. Without direct microdata on training-spell durations, the two are not separately identified with any precision.

We calibrate ϕ from NCES/IPEDS data on average time-to-degree for bachelor's programmes. The national average is approximately 4 years (= 48 months), giving $\phi = 1/48 \approx 0.021$/month. As a robustness check, we also consider $\phi = 1/24$ (2-year associate's degree) to verify that the main results are not sensitive to this choice.

18 Pre-estimated parameter: demographic turnover ν

Rather than calibrating ν from a life-table assumption (e.g. $\nu = 1/480$ for a 40-year working life), we pre-estimate it directly from CPS micro-data. This is feasible because ν maps to an observable labour-force transition: permanent exit from the labour force.

18.1 Why ν cannot be included in the SMM objective

The turnover rate ν appears in the model symmetrically with r: the effective discount rate in most HJBs is $r+\nu$. As a result, ν suffers from the same collinearity: a change in ν can be offset by adjusting r, leaving value functions and hence observable moments essentially unchanged. Beyond this collinearity, ν also governs the inflow rate of new workers (with density $\ell(x)$), which affects the stationary distributions. But the distribution moments (skilled share, training share) are primarily driven by the training cutoff $\bar{x}(z)$ and the type distribution (a_ℓ, b_ℓ), so the marginal effect of ν on these moments is swamped by the effects of the deep parameters that the estimation already targets.

Including ν in the SMM objective alongside r and flow payoffs would make the Hessian near-singular. Pre-estimating ν from a separate data source resolves this without imposing a life-table assumption.

18.2 Construction recipe: CPS labour-force exit hazard

The goal is to measure the monthly rate at which working-age individuals permanently leave the labour force. In the model, ν represents demographic turnover—the rate at which workers exit and are replaced by newborn entrants.

Data source. CPS Basic Monthly, matched panels. Same extract as used for transition rates (Section 2).

Sample.

1. Civilian, age 16–64.
2. In the labour force at date t (employed or unemployed, $\texttt{EMPSTAT} \in \{10, 12, 20, 21, 22\}$).
3. Matched to date $t+1$ using `CPSIDP` (valid MISH pairs: 1–3, 5–7).

Exit indicator. A worker exits the labour force between t and $t+1$ if they are classified as not in the labour force at $t+1$:

$$\text{exit}_{i,t} \;=\; \mathbf{1}\big\{\texttt{LABFORCE}_{i,t} = 2 \;\text{ and }\; \texttt{LABFORCE}_{i,t+1} = 1\big\}.$$

(IPUMS coding: $\texttt{LABFORCE} = 2$ means "in the labour force"; $= 1$ means "not in the labour force".)

Monthly exit hazard.

$$\hat{\nu}_t \;=\; \frac{\displaystyle\sum_{i:\,\text{LF}_t \to \text{NILF}_{t+1}} \texttt{WTFINL}_{i,t}}{\displaystyle\sum_{i:\,\text{LF}_t} \texttt{WTFINL}_{i,t}}.$$

Window average. Average over all valid month-pairs in the pre-FC baseline window (Jan 2003 – Dec 2007):

$$\hat{\nu} \;=\; \frac{1}{T_w} \sum_{t \in w} \hat{\nu}_t.$$

This single number is used throughout all estimation stages.

Why the baseline window. The pre-FC expansion is the "normal" regime. Using the baseline window avoids contamination from crisis-period spikes in discouraged-worker exits (which would bias ν upward). Sensitivity to alternative windows is checked below.

Relation to the model. In the model, workers exit at Poisson rate ν regardless of labour-force status, and each exit is replaced by a newborn entrant drawn from $\ell(x)$. The CPS exit hazard measures $\Pr(\text{NILF}_{t+1} \mid \text{LF}_t)$, which includes both permanent demographic exits (retirement, death, emigration) and temporary exits (returning to school, family leave). In the model all exits are permanent, so the CPS hazard is an upper bound.

To tighten the estimate, one can restrict the sample to ages 50–64 (capturing primarily retirement-driven exits) or ages 16–24 (capturing primarily education-driven exits) and verify that the pooled 16–64 hazard is a reasonable aggregate. Alternatively, the BLS gross-flows data (which tracks LF→NILF flows at the aggregate level) provides a cross-check.

Expected value. The CPS LF→NILF monthly hazard for ages 16–64 is typically in the range 0.03–0.04 (3–4% per month). This is substantially larger than the $1/480 \approx 0.002$ implied by a pure 40-year-working-life assumption, because it includes temporary exits. The appropriate value for the model depends on whether one interprets ν as capturing only demographic turnover (in which case filtering out temporary exits is needed) or all labour-force exits (in which case the raw hazard is correct). The baseline specification uses the raw hazard; a robustness check uses the life-table value as a lower bound.

Standard error. The influence function for $\hat{\nu}$ is $\psi^{\nu}_{i,t} = (\text{exit}_{i,t} - \hat{\nu})/\bar{N}_{\text{LF}}$, following the ratio IF formula. This gives a standard error for the pre-estimated $\hat{\nu}$ that can be incorporated into the SMM inference if desired (treating ν as estimated in a first stage).

19 Sensitivity checks

For each externally set parameter, a targeted sensitivity analysis will be reported:

1. $r \in \{0.0025, 0.0033, 0.0042\}$ (corresponding to 3%, 4%, 5% annual real rates). Re-estimate all deep and regime-specific parameters for each value. Report the change in key outputs (training cutoff, skilled share, welfare).
2. $\phi \in \{1/48, 1/36, 1/24\}$ (4-year, 3-year, and 2-year average programme duration from NCES data). Same protocol.
3. ν: (a) re-estimate using pre-COVID baseline window (2015–2019) instead of pre-FC; (b) use the life-table lower bound $\nu = 1/480$; (c) restrict CPS sample to ages 25–54 to exclude education- and retirement-driven exits. Same protocol.

These checks verify that the main qualitative findings (relative importance of each shock channel, crisis comparison, education-subsidy ranking) are robust to the externally set parameter values.